Protocol for Treating Erectile Dysfunction (ED) Using Low-Intensity Shockwaves

Device: MoreNova - Li50M Dual Linear transducers
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1. Purpose and Scope

This protocol provides a standardized treatment plan for using low-intensity shockwaves to manage erectile dysfunction (ED) of vasculogenic origin. It is intended for healthcare providers using the MoreNova device with the Li50M Dual Linear transducers to target the penile shaft, base, and crura. This protocol applies to men aged 18–80 years diagnosed with vasculogenic ED. For further information on clinical studies and efficacy. Visit our Clinical Data Center.

Device Overview:

MoreNova is a portable Electromagnetic (EM) Low-Intensity Shockwave (LISW) device designed to reduce inflammation, dissolve calcifications, and stimulate angiogenesis, improving outcomes for various conditions in men and women.

2. Introduction

Erectile dysfunction (ED) refers to the inability to achieve or maintain an erection sufficient for sexual intercourse. ED affects approximately 50% of men aged 40–70, with vasculogenic factors being the most common cause.
Low-Intensity Extracorporeal Shockwave Therapy (LiESWT) has become a first-line, non-invasive treatment for vascular ED, promoting angiogenesis, increasing blood flow, and improving endothelial function in the penile tissue.

3. Scientific Review

Current Treatments:

- **Medications**: PDE-5 inhibitors (e.g., sildenafil) are the first-line treatments but may be ineffective for some patients or cause side effects.
- **Mechanical devices**: Vacuum erection devices and penile prostheses are effective but less preferred due to invasiveness or discomfort.
- **Surgery**: Reserved for severe cases but involves risks such as infection and mechanical failure of implants.

Advantages of LiESWT:

- Non-invasive, painless, and safe.
- Clinically proven to improve erectile function even in patients unresponsive to PDE-5 inhibitors.
- Promotes vascular regeneration and increases blood flow in the corpora cavernosa.

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4. Patient Selection Criteria

Inclusion Criteria:

- Diagnosis of vasculogenic ED.
- Age 18–80 years.
- International Index of Erectile Function – Erectile Function Domain Score (IIEF-6) between 11 and 25

Exclusion Criteria:

- Implemented device that is sensitive to electromagnetic radiation.
- History of radical prostatectomy or extensive pelvic surgery
- Pelvic radiation therapy within the last 12 months
- Current or recent cancer diagnosis (within 12 months)
- Neurological or psychiatric conditions that affect erectile function
- Anatomical malformations of the penis
- Venous leak or severe Veno-occlusive dysfunction
- Active infections
- Recent penile surgery
- Chronic hematological diseases or coagulation disorders (INR > 3)

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5. Pre-Treatment Screening

To determine the patient's suitability for LiESWT, conduct the following:

- **Medical and sexual history review**: Assess risk factors such as hypertension, diabetes, and smoking.
- **Physical examination**: Evaluate the penile shaft, base, and crura.
- **Diagnostic imaging**: Use duplex ultrasound or color Doppler to assess arterial flow and rule out severe venous leak.
- **Questionnaires**: Administer the International Index of Erectile Function (IIEF) to evaluate baseline erectile function.

6. Treatment Protocol

Preparation:

- Conduct a thorough medical history and physical examination.
- Counsel the patient about the procedure, benefits, risks, and expected outcomes.

Procedure:

- **Number of Sessions**: 6 sessions
- **Session Frequency**: Twice per week
- **Session Duration**: 20 minutes
- **Applications Per Session**: 2 (penile shaft and crura)
- **Shocks Per Application**: 1,500 (3,000 total shocks per session).

Treatment Parameters:

- **Energy (HV) Settings**: 2.0 kV
- **Frequency (FRQ)**: 2.5 Hz

Post-Treatment Care:

- Advise patients to avoid sexual activity for 24 hours after each session
- Continue PDE-5 inhibitors as prescribed

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7. Treatment Areas and Techniques

The treatment focuses on the following anatomical areas:

- Penile shaft (corpora cavernosa).
- Penile base and root (crura).

Transducers Placement:

- Position the Li50M transducer longitudinally along the penile shaft for uniform energy delivery. (hands free)
- Target the crura from the perineal region. (For more precise targeting, you may use your hands to hold and position the transducers, as demonstrated in the image in sec.11)
- Use ultrasound gel for effective shockwave transmission and patient comfort.

Shocks Distribution:

- Penile shaft: 1,500 shocks. (bilateral)
- Penile base and crura: 1,500 shocks. (bilateral)

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8. Patient Comfort and Safety

To ensure patient comfort and safety during the procedure:

- Position the patient supine for optimal access to the penile shaft and crura.
- Cover the transducer with a new transducer cover for hygiene
- Apply ultrasound gel to the transducer to minimize discomfort and enhance shockwave penetration.
- Monitor patient feedback during the session and adjust energy settings if necessary.

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9. Treatment Outcomes and Follow-Up

Expected Results:

- Improvement in IIEF scores and erectile function within 4–6 weeks after treatment.
- Increased blood flow in the corpora cavernosa observed via ultrasound.

Follow-Up Visits:

- Schedule follow-ups at 4 weeks and 12 weeks post-treatment.
- Reassess erectile function using IIEF scores and duplex ultrasound.

Long-Term Maintenance:

- Consider maintenance sessions every 6–12 months for sustained benefits.

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10. Quality Control

Device Calibration:

- Verify device settings before each session.
- The MoreNova Device recognizes preprogrammed parameters to ensure accuracy.

Clinician Training:

- Ensure all operators are trained in the use of the MoreNova device and Li50M transducer.
- Clinicians should be familiar with this protocol and the "Instructions for Use" provided by the manufacturer.

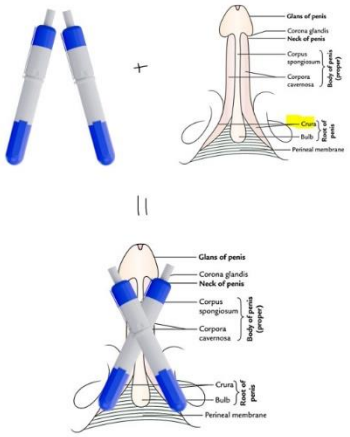
Checklist for Each Session:

- Confirm that the patient meets inclusion criteria and has no contraindications.
- Ensure the device and transducers are clean, sanitized, and functioning correctly.
- Verify transducer placement and shock distribution.
- Fill with cold water and ensure during the treatment that the temperature stays in a proper range.

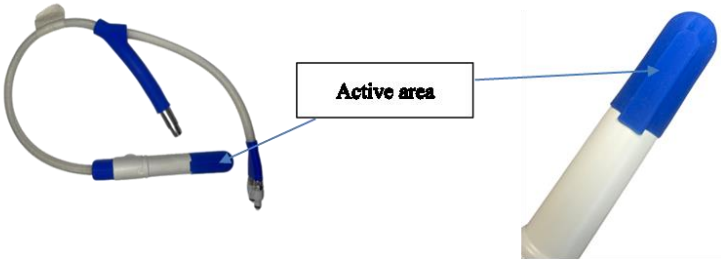
11. Visuals and Diagrams



Step 1 - Penile shaft targeting



Step 2 – Crura targeting



Li50M transducers

12. References

For clinical studies and additional resources, visit the Initia Clinical Data Center.